Two-limb lift_scan in the n = 83 index-calculus pipeline

2026-10-15

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Status (2026-10-15): measured; frozen decision rule applied; verdict `improvement` at both levels. GitHub Actions run 38776966036 (attempt 1, one `ubuntu-24.04` runner, AMD EPYC 9V74, Tier 1 instance scope) ran the frozen protocol. Its artifact `n83-liftscan-38776966036-1` is archived in `run_github_38776966036/`. The one-target online interval on Q0 and on B's 16 fresh points is **1.53 [1.51, 1.53]** and **1.52 [1.51, 1.52]** times faster with `lift_scan`. See Result.

Question

Can an exact, faster least-rotation canonicalizer lower the cost of the current n = 83 index-calculus baseline? We test this at two levels:

- **Stage level.** In the frozen E-1 canonical root stage, does the two-limb binary-lifting `lift_scan` beat variant B, the zero-gap scan? This uses the same frozen pairs as the Tier 1 canonical replay.
- **Online level.** Does it lower the full one-target online interval of the current pipeline on Q0 and on B's 16 fresh points? On these points we also report the distance to the tuned rho.

This is the first step under the new success criterion (AGENTS.md "Success criterion: iterative improvement of the index-calculus baseline", PR #919). That criterion counts every verified, measured improvement to the IC baseline, and makes each one the new baseline.

At n = 131, `lift_scan` beat the zero-gap scan by 1.8–3.8× inside a decomposition-style loop (`canon-n131-loop-20261014`, `canon-n131-20261014`). At n = 83, the Tier 1 replay measured the zero-gap scan at 1.086 [1.076, 1.094] over the full scan on the canonicalization phase (`packed_index_e1_canonical_tier1_20261014`).

Where canonicalization sits in the online path (source audit)

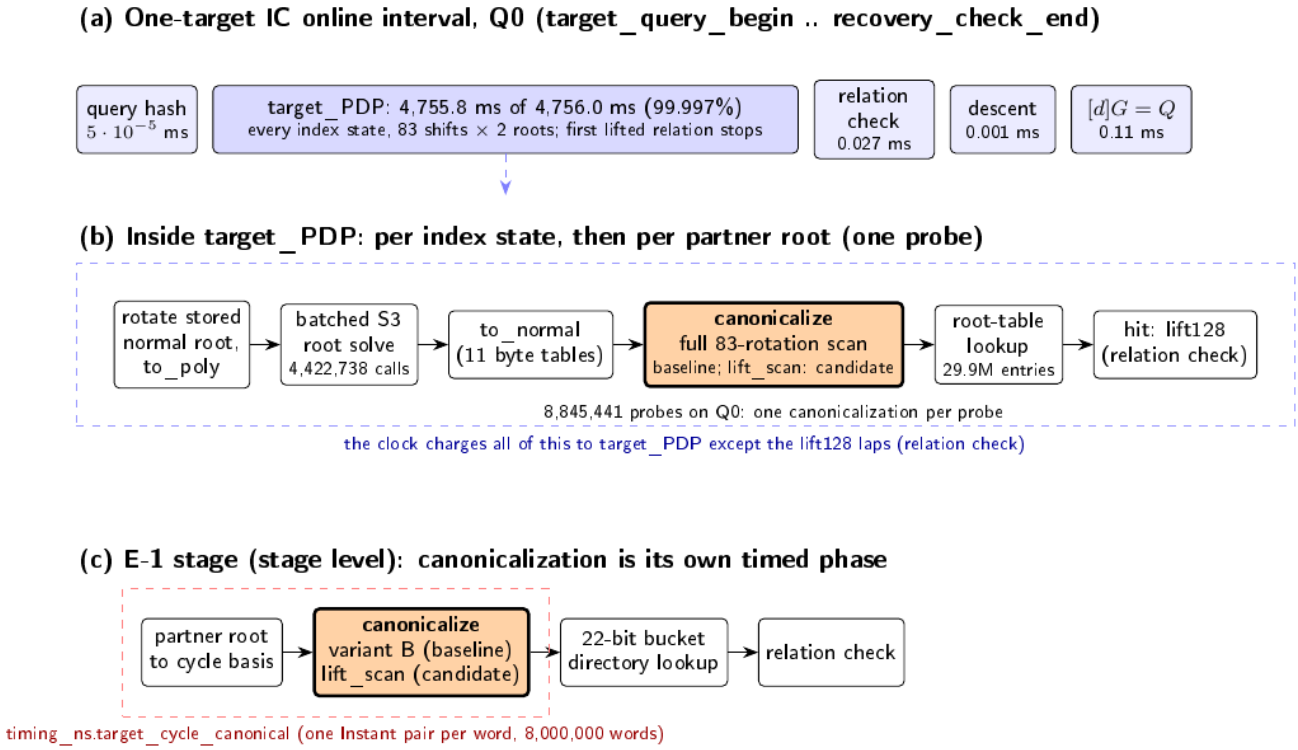


Figure 1: Where canonicalization sits. (a) The one-target online interval and its five phases. (b) The per-probe chain inside target_PDP. (c) The E-1 stage, where canonicalization is its own timed phase. Phase times and counts in (a) and (b) are from reconcile run 38617911080 (`iB770c` on Q0). They are another experiment's figures, shown for location only (source, vector).

- **The canonicalizer in the online path is the full scan, not variant B.** The one-target pipeline is crypto `770f6a69` `examples/koblitz_orbit_dlp_fast_online.rs` plus B's `ic_log_cache.patch`. Its wide path (`mod wide`, n = 83) canonicalizes every partner root in `scan_state128` with `basis.canonical(basis.to_normal.apply(partner))`. That is `NormalBasis128::canonical`, the full 83-rotation scan.
  - The producer also carries a variant-B selector (`canonical_sparse`, `KIC_SPARSE_CANONICAL=1`). It reaches only `canonical_selected` in `build_index128`, which is target-independent setup outside the online interval.
  - Variant B is the canonicalizer of the E-1 packed-index **stage** only (`native_packed_index.rs`, `packed_bucket22_fastcanon`). It is not the canonicalizer of the one-target online path.
  - The online baseline arm is therefore the full scan. Variant B online (`qG`) is a secondary arm.
- **Its place in the interval.** The online interval is the producer's own clock, from `target_query_begin` to `recovery_check_end`, with five exclusive phases. Canonicalization is inside `target_PDP`. It runs once per probe, after the batched S3 root solve and the `to_normal` byte-table conversion, and before the root-table lookup. The clock does not time it separately.
- **Its share of the interval.** In reconcile run 38617911080 (`iB770c` on Q0), `target_PDP` was 4,755.8 of 4,756.0 ms (99.997%), over 8,845,441 probes and 4,422,738 S3 root calls with 1 relation check. That is another experiment's figure; it locates canonicalization but does not size it. This run measures the share directly:
  - The planted arm `qFx2` does one more, discarded, full scan per probe. So `(qFx2 - qF)/qF` is the marginal share of one canonicalization in the baseline interval, with a bootstrap interval.
  - `(qF - qL)/qF` is the saving `lift_scan` realizes.
  - The `target_pdp_ms / online_ms` share is reported too.
  - Measured by this run: see Result.

The canonicalizer

lift2/src/lib.rs implements `L2<N, B>`. It holds an `N`-bit cyclic word in two `u64` limbs, `lo` with bits `0..B-1` and `hi` with bits `B..N-1`. The production geometry is `N = 83`, `B = 64`, so the limbs hold 64 and 19 bits. It is the variant-A analogue of canon-n131's `lift_scan`, cut down from three limbs to two:

- `Z_1 = !x`, and `Z_{2a} = Z_a & rotl(Z_a, a)`, for runs of length 1, 2, 4, ..., 64.
- Greedy top-down lifting gives the tops of all longest cyclic zero runs. It uses only constant rotations and mask selects, behind an `asm!` value barrier.
- A loop over those tops then keeps the least rotation, with ties going to the smaller shift.
- The words `0` and `mask` return shift 0, as the full scan does.

The same source file is compiled into the stage binary and into the online producer through `#[path]` modules. The online producer monomorphizes it per wide degree, so every rotation shift is a constant.

Exactness (verified locally; the CI run repeats all of it before timing)

Check	Size	Disagreements
`lift2` tests: every word at N = 3..20 (2,097,144 words), each under every two-limb split $\lceil N/2 \rceil \le B < N$ for $N \le 16$ and 3–4 splits for N = 17..20, composite N included (periodic and tied words)	7,252,648 (word, split) cases	0
N = 83: 0, all ones, every weight-1 and weight-82 word	168	0
N = 83: one zero run of every length 1..82 at every top position (± random background)	13,612	0
N = 83: longest runs across the limb boundary (bits 58–70, 60–64, 63–64, 64–66), ending at bit 82, at bit 0, and wrapping 82→0	1,446	0
N = 83: two- and three-way ties of the longest run	200,000	0
N = 83: fixed-seed words of mixed density	2,000,000	0
every dispatched wide degree (14 degrees, 65 ... 127) on fixed-seed and edge words	14 × 50,005	0
stage `packed_bucket22_verifycanon`: every word the frozen E-1 stage canonicalizes (4,000,000 index + 8,000,000 target + 64 sample), full vs variant B vs `lift_scan`	12,000,064	0 and 0
stage pair certificate (20 fields incl. lookup-sample keys and shifts), modes `liftcanon`, `fastcanon_x2`, `verifycanon`	3 runs	equals frozen `7cfba635...acaff8`

Check	Size	Disagreements
online `KIC_TARGET_CANON=verify`: every probe word of the one-target scan, full vs variant B vs `lift_scan`	27,225,926 (Q0, w00, w01, w02 locally; all 17 points on the runner)	0 and 0

The test can fail. Two mutations were tried:

- reversing the tie-break: 11 of 24 tests fail;
- zeroing the cross-limb carry in `shl`: 22 of 24 fail.

A third mutation removed the last lifting stage. It still passes, and should: the candidate set then becomes a superset of the longest-run tops, which is slower but still exact.

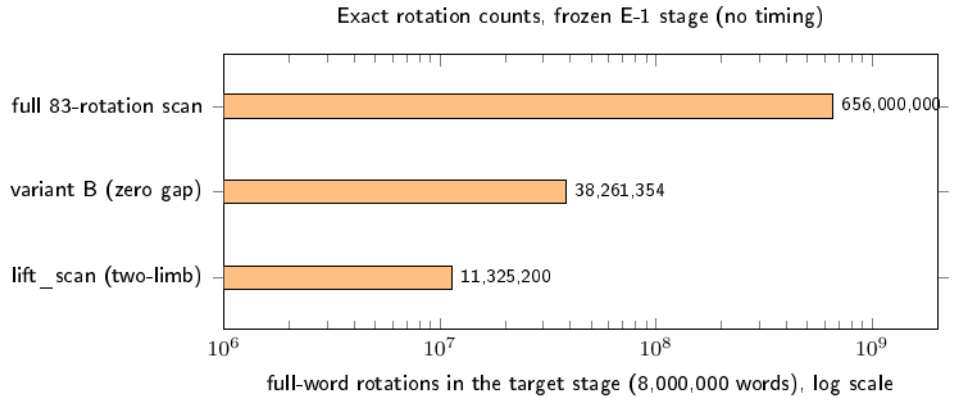


Figure 2: Exact rotation counts in the frozen E-1 target stage (8,000,000 words). Full scan and variant B are the frozen E-1 rows. The lift_scan count is the number of nonzero-shift candidate rotations, from the local exactness run; the 13 constant rotations of the lifting are not counted. These are counts, not times (source, vector).

Frozen design

`protocol.json` and `schedule.json` were written and committed before any timing. `make_schedule.py --check` and `restore.py` re-verify them, and every source digest, on the runner.

Arm	Binary / mode	Role
`sB`	stage `packed_bucket22_fastcanon`	stage baseline: variant B
`sA`	stage `packed_bucket22_liftcanon`	stage candidate: `lift_scan`
`sBaa`	stage `packed_bucket22_fastcanon`	stage A/A control
`sBx2`	stage `packed_bucket22_fastcanon_x2`	stage planted positive control: one more, discarded, variant-B call per word
`sV`	stage `packed_bucket22_verifycanon`	exactness, 12,000,064 words, not timed
`qF` / `fF`	online `KIC_TARGET_CANON=full`	online baseline: the pipeline's full scan, unchanged behaviour
`qL` / `fL`	online `lift`	online candidate
`qG`	online `gap`	secondary: variant B in the online path
`qFaa`	online `full`	online A/A control

Arm	Binary / mode	Role
qFx2	online full_x2	planted positive control; also sizes one canonicalization per probe
qV	online verify	exactness on Q0 and every fresh point, not timed
rT	A's tuned rho (Fold/64/DP6, crypto 770f6a69)	distance to the strongest n = 83 rho measured so far (reconcile rT770)

- **Stage panel.** The frozen E-1 workload 981ac530e67e on curve EC1N83Ckb1h876c2921cb64, with 2,000,000 pair states and a 1,024 MiB RSS cap. The binary is the frozen source with one change, the canonicalizer behind Cycle::canonical. There are 16 blocks of the four timed arms in seeded shuffled order.
  - Every arm must reproduce the frozen E-1 pair certificate and its mode's frozen rotation counts.
  - Primary: the canonicalization phase timing_ns.target_cycle_canonical, as the per-block ratio sB/sA. The estimate is the median of the 16, with a paired bootstrap (10,000 resamples; seed 20261011; indices 250 and 9749).
  - Secondary: the target stage.
- **Online panels.** Curve EC1N83Ce1h35c7ced6b497, with the retained K600 base d4a4da26...a9f7. B's archived factor-log table is re-verified in the pinned oracle before timing, and again in every process.
  - Q0 has 8 blocks: qF, qL, qG, qFaa and qFx2, plus rT at seed 202610060 + block. The primary is the median of the per-block qF/qL ratios.
  - B's 16 fresh points (seeds 0..15) each get fF, fL and rT at seed 830000 + w. The primary is the section-23 ratio of means, bootstrapped over points.
- **Order and isolation.** Exactness comes first, then the stage blocks, then Q1, F1, ..., Q8, F8. The whole run is one GitHub-hosted ubuntu-24.04 runner. Every process is pinned to one CPU, with the isolated_bench Tier 1 counters and memory gates on each.
- **Gates and controls.** All identity checks must hold on every point:
  - the same probes, relation and scalar in every IC arm;
  - IC and rho agreeing on the scalar;
  - an oracle replay of [d]G = Q;
  - Q0's scalar equal to the sidecar.

  The A/A intervals must lie inside [0.97, 1.03], and the planted lower bounds must exceed 1. Otherwise that level is invalid.
- **Decision.** A lower bound above 1 is improvement; an upper bound below 1 is regression; otherwise not_separated.
  - The online level is an improvement only when both Q0 and the fresh section-23 primary are improvements.
  - An online improvement makes the lift_scan pipeline the new one-target baseline (the ratchet). A stage-only win is recorded as a stage improvement.
- **S.** S_probe = probes / sqrt(r) is unchanged by construction, because probes are a pair field. S_rho=(IConline/tuned-rhomsperstep)/sqrt(r) prices the online interval in this job's rho steps. It is reported for the old and the new baseline, with its delta. Here $\sqrt{r}$ = 92,573,139.2 and r = 8,569,786,107,849,059.
- **Runtime estimate.** About 1.5–2.3 h of panel time after a 15–25 min build. That rests on the reconcile run, where the cached IC process wall was 41–45 s and the tuned rho took 2–3 s. The time guard writes skipped_time_guard rows rather than dropping them.

Result

Run 38776966036, attempt 1, sha 37dcdfaf8 (run_github_38776966036/panel/result.json and result_table.md). verify_receipt.py re-run on a copy of the archived panel, with the pinned crypto 770f6a69 verifier sources, reproduced result.json byte for byte. Intervals are 95% bootstrap (frozen).

Gates. All hold: every scheduled row valid; stage exactness (12,000,064 words, 0 disagreements) and online exactness (every probe word on Q0 and the 16 fresh points); stage pair certificate equals the frozen one; probes, relation and scalar identical across arms per point; IC and tuned rho agree on the scalar; every point replays as [d]G = Q; Q0 equals the validation sidecar. Both A/A controls lie inside [0.97, 1.03] and both planted controls exceed 1. (Only the untimed qV exactness rows carry a major page faults contention flag.)

Frozen verdict. Online: **improvement** (Q0 and the section-23 fresh primary are both improvements). Stage:

improvement. Ratchet: **the `lift_scan` pipeline is the new one-target baseline.**

statistic	estimate [95%]	reading
Online Q0 primary, full scan / lift_scan (median of 8 blocks)	**1.531 [1.509, 1.534]**	improvement
Online fresh primary (section 23, ratio of means over 16 points)	**1.515 [1.506, 1.525]**	improvement
Fresh median per-point ratio	1.518 [1.511, 1.528]	
Q0 variant B (gap scan) / lift_scan	1.000 [0.997, 1.012]	tie
Q0 full scan / variant B	1.518 [1.510, 1.533]	
Online A/A, full / full	1.000 [0.992, 1.006]	pass
Online planted, full + discarded full / full	1.408 [1.401, 1.415]	pass
Canonicalization share of the Q0 interval	0.408 [0.401, 0.415]	derived
Realized saving on Q0, (qF - qL)/qF	0.347	
Stage primary, canonicalization phase B / lift_scan	**2.272 [2.268, 2.275]**	improvement
Stage secondary, target stage B / lift_scan	1.155 [1.140, 1.169]	improvement
Stage A/A (canonicalization; target stage)	1.000 [0.998, 1.002]; 1.001 [0.989, 1.022]	pass
Stage planted, B + discarded B / B	1.760 [1.755, 1.763]	pass

Probes are unchanged by construction (8,845,441 on Q0), so the saving is per probe: Q0 ns/probe 480.3 (full scan) to 316.0 (`lift_scan`); variant B in the online path is 314.7.

New baseline `S` and its delta. `S_probe = probes/sqrt(r)` is unchanged (Q0 0.09555, fresh 0.04299). `S_rho` is the online interval in this job's tuned-rho steps over sqrt(r) (rho 245.5 ns/step):

panel	`S_rho` old (full scan)	`S_rho` new (`lift_scan`)	delta
Q0 (online mean 4,248 ms to 2,795 ms)	0.1869	0.1230	**-0.0639** (-34%)
16 fresh (1,903 ms to 1,256 ms)	0.0837	0.0552	**-0.0285** (-34%)

Gap to the tuned rho (rT770, Fold/64/DP6), rho online / IC online. Above 1 means the IC is faster; below 1, rho is faster.

panel	old baseline	new baseline
Q0, ratio of means (8 blocks)	0.597 [0.442, 0.758]: rho faster	**0.911 [0.673, 1.164]: not separated**
Q0, median of block ratios	0.552 [0.342, 0.917]	0.838 [0.516, 1.405]
16 fresh, section 23	0.984 [0.476, 3.408]	1.491 [0.727, 5.145]

So the change closes the Q0 gap from rho being 1.68 times faster (separated) to parity within noise. On the 16 fresh points the gap was never separated and is not now; those intervals are wide because three points carry most of the probes. No gap to Bernstein-Lange was measured here (this protocol has no BL arm).

Reading.

- The improvement is a measured engineering change to the online stage on this instance. It is exact: probes, relations and scalars are identical to the full-scan baseline on all 17 points.

- The saving matches the planted control: one extra full scan per probe costs 40.8% of the interval and `lift_scan` removes 34.7% of it, close to variant B, which is the same within noise (1.000 [0.997, 1.012]).
- The stage ratio (2.27) is larger than the target-stage ratio (1.155) because the canonicalization phase is a fraction of that stage.
- Not claimed: a gain at other sizes, a cold or setup gain (the index build still uses the full scan), or any statement about beating rho.

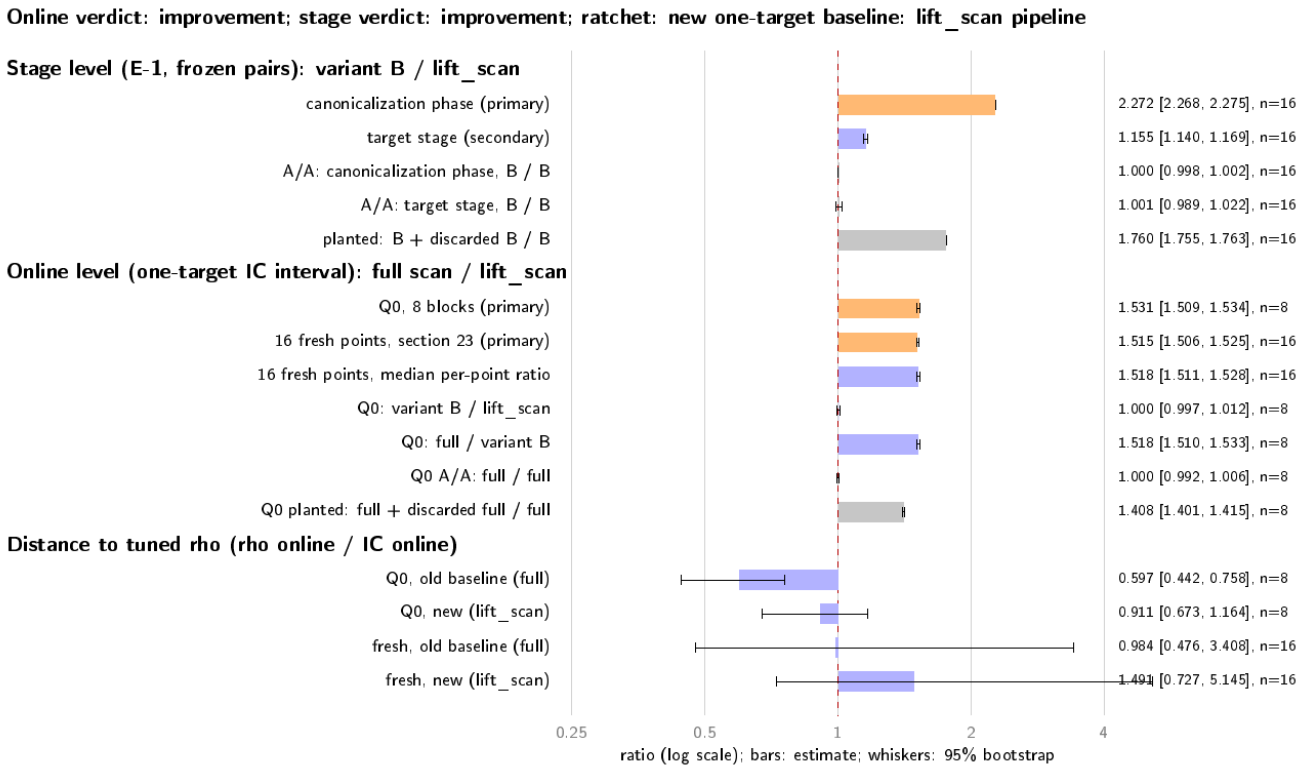


Figure 3: Result summary

Figure. Every frozen ratio with its 95% bootstrap interval, and the distance to the tuned rho (source, vector, rendered by `make_figures.py`).

Local validation (2026-10-15, cloud agent VM; no timing claims)

The VM is a 4-vCPU Firecracker guest shared with other agents. Its clocks are not evidence, and no time from it is reported.

- **Build and tests.**
  - `lift2`: 24 tests passed.
  - stage crate: 5 tests passed. These include the frozen `zero_gap_canonical_matches_full_rotation_scan` and the new `lift_planted_and_verify_modes_match_full_scan_at_n83`.
  - online crate: the producer's own 3 tests and the tuned rho's 4 tests passed.
  - Toolchain: rustc 1.97.0 here. The runner pins 1.99.0, as the reconcile run did.
- **Exactness.** See the table above. The local smoke receipt is in `smoke/`, with correctness only: `correctness.json` holds no timing values.
- **Smoke run.** `run_schedule.py--segmentsX,S1,Q1,F1--max-runs-per-segment6` ran the real binaries through the real wrapper, gates and verifier. A container restart cut it short after the five exactness rows. A second part then ran `--segmentsS1,Q1--one-per-arm`. All 15 rows were valid. Every IC arm on Q0 found the same 8,845,441 probes, the same relation and the scalar 3141592653589793, which equals the sidecar. The tuned rho agreed. A full dry run of the analysis and figure scripts on synthetic timings worked end to end; its outputs were discarded.

Files

File	Role
protocol.json, schedule.json, make_schedule.py	frozen protocol and run order
lift2/	the two-limb lift_scan crate and its exhaustive tests
stage/	E-1 stage binary (frozen source plus the canonicalizer selector), frozen Cargo.lock
online/, make_online_source.py	online producer derived from B's producer by counted exact replacements, plus A's tuned rho; reconcile's Cargo.lock
restore.py, check_log_table.py	restore and digest-check both crypto revisions and all local sources; re-verify the factor-log table
arm.py, run_schedule.py	one arm per pinned process; the schedule runner (reconcile's, adapted)
verify_receipt.py	gates, statistics, decision rule; result.json, result_table.md
make_figures.py, figures/	figure sources (TikZ), PDF and PNG renders
.github/workflows/n83-liftscan.yml	the Tier 1 job (push to claude/n83-liftscan; receipts, README, figures and figure scripts excluded from its path filter)
README.pdf	this report: pandocREADME.md-oREADME.pdf--pdf-engine=pdflatex-Hreport_header.tex--lua-filterreport_filter.lua-Vgeometry:margin=2cm-Vfontsize=10pt-Vcolorlinks=true

Limits

- The verdicts are Tier 1 and instance-scoped: one runner, one pinned CPU, not host-isolated.
- Q0's intervals cover repeat and timing variation, not target variation. The fresh panel carries the target variation.
- The stage's canonicalization phase brackets every word with an Instant pair. That overhead is common to both arms, and it dilutes the stage ratio.
- The online ratio is the improvement that counts for the baseline.
- The index build still uses the full scan, because it is target-independent setup. Moving it to lift_scan would change setup cost, not the online interval. It is the obvious next step for end-to-end cold S.
- The tuned-rho gap is the distance remaining. It is not a pass/fail gate, and nothing here is a claim to beat rho.